

Eleven Primitives and Three Gates: The Universal Structure of Computational Imaging

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Abstract

Every computational imaging system—from a coded-aperture spectral camera to an MRI scanner to a cryo-electron microscope—converts physical reality into measurements through a chain of physical transformations. Despite the apparent diversity of these systems across carrier families, length scales, and modalities, here we show that the space of all such forward models has *finite structural complexity*. We prove the **Finite Primitive Basis Theorem**: every imaging forward model in a broad operator class admits an ε -approximate representation as a directed acyclic graph over exactly 11 physically typed primitives—a basis that is both sufficient and minimal¹. This is a discovery, not a design choice: the primitive library saturates at 11 as modality coverage grows past 30 registered systems, with no new primitive required for the most recent 18 additions, proving that the structural complexity of imaging physics is bounded. We further establish the **Triad Decomposition**, a universal diagnostic law governing every failure in computational image recovery: reconstruction error decomposes into exactly three independent root causes—information deficiency, carrier noise, and operator mismatch—mirroring the role of entropy, thermal noise, and free-energy barriers in the thermodynamics of irreversible processes. Hardware validation across twelve modalities spanning all five carrier families—photons, electrons, X-rays, nuclear spins, and acoustic waves—including cryo-electron microscopy and clinical cone-beam CT, confirms that operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck under standard operating conditions, recoverable by +0.8 to +10.7 dB (± 0.3 dB, 95% bootstrap CI) through forward-model correction alone—often exceeding the gain from upgrading a classical solver to a state-of-the-art deep network. The framework’s scope extends from coded aperture cameras to clinical CBCT scanners to cryo-EM instruments used for drug discovery, demonstrating that the structural bottleneck in image recovery is universal across scientific imaging. These results shift the primary locus of investment in computational imaging from solver design to calibration, and provide a unifying blueprint for decomposing forward models across physical measurement science.

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Introduction

Every imaging instrument—whether it captures spectra, spins, X-rays, or sound waves—converts a physical scene into digital measurements through a chain of transformations: a wave propagates, interacts with the object, passes through optics, and is recorded by a detector. Computational imaging reconstructs the scene by inverting this chain. The reconstruction algorithm assumes a mathematical model of the chain (the “forward model”), but this model inevitably diverges from reality: optics shift, detectors drift, and calibration degrades. When the model is wrong, the reconstruction fails—not because the algorithm is weak, but because it is solving the wrong problem.

This paper establishes that the space of imaging forward models has a surprisingly simple and universal structure, and that exploiting this structure reveals operator mismatch as the systematic, cross-modality bottleneck in reconstruction quality. Over the past decade, the community has invested enormous effort in improving reconstruction algorithms—progressing from compressed sensing^{2,3} and plug-and-play priors⁴ to deep unrolling networks⁵ and vision transformers⁶—yet these algorithms routinely fail when deployed on real instruments⁷. Calibration methods exist for specific instruments—ESPIRiT⁸ for MRI, ePIE⁹ for ptychography—but they do not generalise across modalities, and no systematic framework decomposes reconstruction failures into root causes.

Here we establish two complementary foundations within Physics World Models (PWM). The **Finite Primitive Basis Theorem** (Theorem 3) proves that every imaging forward model in a broad operator class $\mathcal{C}_{\text{img}}$ admits an ε -approximate representation as a typed directed acyclic graph (DAG) over exactly 11 canonical primitives—and that this library is minimal¹. The **Triad Decomposition** proves that every reconstruction failure decomposes into three gates—information deficiency (**Gate 1**), carrier noise (**Gate 2**), and operator mismatch (**Gate 3**)—with a formal condition (Theorem 5) under which **Gate 3** dominates. The 11 primitives *enable* the 3-gate diagnosis: the DAG structure localizes mismatch to a specific physical operator, making correction actionable.

We validate both results across twelve modalities spanning all five carrier families—optical photons (CASSI¹⁰, CACTI¹¹, SPC¹², lensless, compressive holography, fluorescence microscopy), X-ray photons (CT¹³, cone-beam CT), electrons (ptychography¹⁴, cryo-EM), nuclear spins (MRI^{15,16}), and acoustic waves (ultrasound)—with hardware validation on ten modalities confirming Triad predictions on physical measurements. The validated modalities include two Nobel Prize-winning techniques (cryo-EM, 2017 Chemistry; super-resolution fluorescence, 2014 Chemistry), three clinically deployed instruments (CT, CBCT, MRI), and the first acoustic-carrier validation in computational imaging. A held-out closure test on 8 additional modalities confirms basis completeness with saturating growth. The proof of Theorem 3 is in Supplementary Note 12; formal primitive semantics and typed DAG denotation are developed in¹.

71 **Why this matters beyond computational imaging.** Any physical measurement system—
72 seismographs, telescopes, particle detectors, environmental sensor networks—faces the same
73 structural problem: a forward model maps the quantity of interest to observations, and
74 model fidelity limits what can be recovered. The finding that eleven primitives span the
75 entire space of imaging forward models raises a fundamental question about the grammar of
76 physical measurement: if five carrier families, twelve validated modalities, and thirty regis-
77 tered systems all compose from the same small set of operators, is this a contingent feature
78 of imaging, or does it reflect a deeper constraint on how information flows from physical
79 systems to digital observations? The OPERATORGRAPH formalism provides a template
80 for decomposing forward models in other domains into typed, composable primitives with
81 validated adjoints. The Triad Decomposition—separating information capacity, noise, and
82 model fidelity as independent failure modes—applies wherever an inverse problem is solved
83 computationally. The practical lesson is domain-general: before investing in a better solver,
84 diagnose whether the model is the bottleneck.

85 The Finite Primitive Basis

86 A foundational question in computational imaging is whether the diversity of imaging for-
87 ward models—from coded aperture cameras to MRI scanners to electron microscopes—can
88 be captured by a small, fixed set of primitive operators. We answer affirmatively.

89 **The OperatorGraph representation.** Every imaging forward model is encoded as a
90 typed directed acyclic graph (DAG) in which each node wraps a single primitive physical
91 operator and edges define the data flow from source to detector. Every primitive implements
92 both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring
93 $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision. Each edge carries tensor shape and
94 dtype metadata, enabling static validation before execution. We call this formalism the
95 OPERATORGRAPH intermediate representation (IR).

96 **Eleven canonical primitives.** The OPERATORGRAPH IR defines a library of exactly 11
97 canonical primitives $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$:
98 The Detect nonlinearity η is restricted to five canonical families (linear-field: $\eta(x) = gx$;
99 logarithmic; sigmoid; intensity-square-law: $\eta(x) = g|x|^2$; coherent-field), each with at most
100 2 scalar parameters. The Transform primitive Λ applies a pointwise nonlinear function
101 within the physics chain (not at the detector) and is restricted to five families: exponential
102 attenuation ($e^{-\alpha x}$, Beer–Lambert), logarithmic compression, phase wrapping ($\arg(e^{ix})$),
103 polynomial ($\sum a_k x^k$, $d \leq 5$), and saturation. Both Detect and Transform are pointwise
104 and parameter-limited, preventing either from becoming a universal approximator (see¹ for
105 formal definitions and non-universality proofs).

#	Primitive	Notation	Physical action
1	Propagate	$P(d, \lambda)$	Free-space wave propagation
2	Modulate	$M(\mathbf{m})$	Element-wise multiplication (mask, coil, absorption)
3	Project	$\Pi(\theta)$	Radon line-integral projection
4	Encode	$F(\mathbf{k})$	Fourier-domain encoding (k -space)
5	Convolve	$C(\mathbf{h})$	Spatial convolution (PSF)
6	Accumulate	Σ	Summation over spectral/temporal axis
7	Detect	$D(g, \eta)$	Detector response (5 canonical families)
8	Sample	$S(\Omega)$	Sub-sampling on index set Ω
9	Disperse	$W(\alpha, a)$	Wavelength-dependent spatial shift
10	Scatter	$R(\sigma, \Delta\epsilon)$	Direction change and/or energy shift
11	Transform	$\Lambda(f, \boldsymbol{\theta})$	Pointwise nonlinear physics operation

Physics-stage mapping. Each factor of a forward model is classified into one of six physics-stage families and mapped to primitives from $\mathcal{B}$: propagation $\rightarrow \{P, C\}$; elastic interaction $\rightarrow \{M\}$; inelastic interaction (scattering) $\rightarrow \{R\}$; pointwise nonlinear physics $\rightarrow \{\Lambda\}$; encoding–projection $\rightarrow \{\Pi, F\}$; detection–readout $\rightarrow \{\Sigma, S, W, C, D\}$.

Fidelity levels. Forward models can be written at increasing physical fidelity: linear shift-invariant (simplest), linear shift-variant, nonlinear ray/wave-based, and full-wave or Monte Carlo. All 11 primitives apply uniformly across every fidelity level. Linear stages are composed from the first 10 primitives; nonlinear stages (beam hardening, phase wrapping, saturation) are captured by Transform Λ . The theorem therefore covers the full fidelity spectrum without restriction.

Five physical carriers. The library spans five carrier families—photons, electrons, spins, acoustic waves, and particles—with 26 registered modality templates, of which 12 now have full end-to-end correction validation across all five carrier families (Extended Data Table 2; Supplementary Table S3).

Theorem statement.

Definition 1 (Imaging Operator Class). $\mathcal{C}_{\text{img}}$ consists of all imaging forward models expressible as a finite sequential-parallel composition of stages—each either linear with bounded operator norm ($\|H_k\| \leq B$) or pointwise nonlinear with bounded Lipschitz constant ($\text{Lip}(\Lambda_k) \leq L$)—with stage count $K \leq N_{\text{max}}$.

Definition 2 (ϵ -Approximate Representation). A typed DAG G with nodes $V \subseteq \mathcal{B}$ is an ϵ -approximate representation of $H \in \mathcal{C}_{\text{img}}$ if $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \epsilon$, where $\delta > 0$ is a regularization constant, and $|V| \leq N_{\text{max}}$, $\text{depth}(G) \leq D_{\text{max}}$.

Theorem 3 (Finite Primitive Basis). For every $H \in \mathcal{C}_{\text{img}}$, there exists a typed DAG G with $V \subseteq \mathcal{B}$ that is an ϵ -approximate representation of H .

130 *Proof sketch.* By Definition 1, $H = H_K \circ \dots \circ H_1$ with $K \leq N_{\max}$. Each factor is classified
 131 into one of six physics-stage families and realized by primitives: propagation $\rightarrow P$ or C ; elas-
 132 tic interaction $\rightarrow M$; inelastic interaction $\rightarrow R$; pointwise nonlinear physics $\rightarrow \Lambda$; encoding-
 133 projection $\rightarrow \Pi$ or F ; detection-readout $\rightarrow$ finite composition from $\{\Sigma, S, W, C, D\}$. For
 134 linear stages, per-factor errors compose via sub-multiplicativity: $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot$
 135 B^{K-1} . For nonlinear stages, the Lipschitz composition bound $\|H(\mathbf{x}) - \hat{H}(\mathbf{x})\| \leq \sum_k \varepsilon_k \prod_{j>k} \gamma_j$
 136 applies, where γ_j is the Lipschitz constant of stage j . Both bounds yield $\leq \varepsilon$ under the
 137 stated regularity conditions. Full proof in Supplementary Note 12; formal primitive seman-
 138 tics and typed DAG denotation in¹. $\square$

139 **Scope of $\mathcal{C}_{\text{img}}$.** Covers: all clinical, scientific, and industrial imaging modalities—linear and
 nonlinear—including coded aperture, interferometric, projection, Fourier-encoded, acoustic,
 scattering, beam hardening, phase wrapping, multiple scattering, Monte Carlo systems, quan-
 tum state tomography, and relativistic systems. No modality is excluded. Formal definitions
 and proofs in¹.

140 **Extension protocol.** A new primitive is warranted when no DAG over $\mathcal{B}$ achieves $e_{\text{img}} \leq$
 141 ε within the complexity bounds. The extension requires: (1) validated forward/adjoint,
 142 (2) demonstrated representation gap, (3) error reduction below ε , (4) need by ≥ 2 modali-
 143 ties, (5) backward-compatible closure re-test. Scatter (R) was added via this protocol when
 144 Compton imaging produced $e_{\text{img}} = 0.34$ without it.

145 **Held-out closure test.** To validate Theorem 3 empirically, we conduct a closure test un-
 146 der a frozen protocol: the primitive library $\mathcal{B}$ (11 primitives), Detect families (5), Transform
 147 families (5), fidelity threshold ($\varepsilon = 0.01$), and complexity bounds ($N_{\max} = 20$, $D_{\max} = 10$)
 148 are all frozen before evaluation. We test 5 held-out modalities (OCT, photoacoustic, SIM,
 149 phase-contrast X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imag-
 150 ing, THz-TDS, Compton scatter imaging):

Modality	e_{img}	#Nodes/Depth	Triad transfer	New prim.?
OCT	< 0.01	5 / 3	Y	N
Photoacoustic	< 0.01	4 / 3	Y	N
SIM	< 0.01	4 / 3	Y	N
Phase-contrast X-ray	< 0.01	5 / 4	Y	N
Electron ptycho	< 0.01	4 / 3	Y	N
Ghost imaging	< 0.01	4 / 3	Y	N
THz-TDS	< 0.01	3 / 2	Y	N
Compton scatter	$< 0.01^*$	4 / 3	Y	Y (R)

*With R in library; $e_{\text{img}} > 0.01$ without R .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds $N_{\max} = 20$, $D_{\max} = 10$).

151 Seven of eight modalities decompose with the frozen library. Quantum ghost imaging
 152 is operator-equivalent to a single-pixel camera at the image-formation level—sharing the
 153 canonical DAG Source $\rightarrow M \rightarrow \Sigma \rightarrow D$ despite fundamentally different source physics.

THz-TDS requires only the coherent-field Detect family. Compton scatter imaging triggered the extension protocol (Section “Extension protocol” above): its representation gap ($e_{\text{img}} = 0.34 \gg \varepsilon$) met the falsifiable criterion for basis incompleteness, and the disciplined resolution—adding Scatter (R)—covers 5+ additional modalities (Raman, fluorescence, DOT, Brillouin) while preserving all existing decompositions. The basis grows from 9 to 11 (22% increase) while modality coverage grows by 19%+.

Basis-growth saturation. Plotting the number of distinct primitives K against the number of registered modalities N reveals clear saturation (Figure 7): 9 of 11 primitives are introduced by the first 10 modalities, Disperse (W) by CASSI-type spectral systems, Scatter (R) when Compton/Raman-class modalities enter, and Transform (Λ) when nonlinear physics stages (beam hardening, phase wrapping) are encountered. The growth is sublinear and saturating: $K = 11$ at $N = 30$, with no new primitive required for the most recent 18 modalities added. This saturation is consistent with Theorem 3: once all six physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

Theorem tightness and minimality. The theoretical complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$) are conservative: empirically, all 26 registered and 8 held-out modalities (~ 30 unique, accounting for 4 overlaps) require ≤ 6 nodes and depth ≤ 5 (held-out closure test table above), with the median modality using 4 nodes at depth 3. The gap between empirical and theoretical bounds reflects the compactness of real physical imaging chains. Moreover, the basis is *minimal*: for each of the 11 primitives, we exhibit a witness modality whose representation error exceeds ε when that primitive is removed (Supplementary Note 10, Proposition 1 therein; beam hardening CT witnesses the necessity of Transform Λ). Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds.

The Triad Decomposition

The TRIAD DECOMPOSITION asserts that every failure in computational image recovery within $\mathcal{C}_{\text{img}}$ can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

Why exactly three gates. The reconstruction pipeline has exactly three inputs: a sensing geometry H that determines what information is captured (Gate 1), a physical carrier whose statistics determine the noise floor (Gate 2), and a forward model H_{nom} that the solver inverts (Gate 3). Any reconstruction error must trace to one of these

188 inputs: information that was never captured (null space of H), information that was cap-
 189 tured but corrupted (carrier noise), or information that was captured but misinterpreted
 190 (model error $H_{\text{nom}} \neq H_{\text{true}}$). This trichotomy is exhaustive: the MSE decomposes as
 191 $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$ (Supplementary Note 1), with no residual term.
 192 Solver suboptimality (e.g., insufficient iterations or regularization mismatch) is subsumed
 193 by Gate 3 in the Triad formalism, because the solver’s implicit forward model includes its
 194 regularization assumptions.

195 **Gate 1: Recoverability.** **Gate 1** asks whether the measurement encodes sufficient in-
 196 formation about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps
 197 the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$
 198 defines signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is
 199 large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the
 200 measurement design and can only be remedied by acquiring additional data.

201 **Gate 2: Carrier Budget.** **Gate 2** asks whether the signal-to-noise ratio (SNR) is suffi-
 202 cient. Every physical carrier—photons, electrons, spins, acoustic waves, particles—is sub-
 203 ject to fundamental noise limits: shot noise for photon-counting systems, thermal noise
 204 in electronic detectors, T_1/T_2 relaxation noise in magnetic resonance. When the carrier
 205 budget is too low, the measurement is dominated by noise and the reconstruction degrades
 206 regardless of operator fidelity.

207 **Gate 3: Operator Mismatch.** **Gate 3** asks whether the forward model assumed by the
 208 solver matches the true physics. The solver operates with a nominal operator H_{nom} , but
 209 the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction
 210 targets a phantom inverse problem. **Gate 3** failures are insidious because they produce
 211 structured artifacts that mimic signal content, leading practitioners to blame the solver
 212 rather than the model. Sources include geometric misalignment (mask shift, rotation),
 213 parameter drift (coil sensitivity variation, gain instability), and model simplification.

214 **Definition 4** (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{img}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and*
 215 *solver using nominal operator H_{nom} . The reconstruction error decomposes as $\text{MSE} \leq$*
 216 *$\text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$, where $\text{MSE}^{(G1)}$ is the null-space loss (Supplementary*
 217 *Eq. S1), $\text{MSE}^{(G2)}$ is the noise-floor term (Supplementary Eq. S3), and $\text{MSE}^{(G3)}$ is the*
 218 *mismatch-induced term (Supplementary Eq. S5).*

219 **Theorem 5** (Gate 3 Dominance). *For any $H \in \mathcal{C}_{\text{img}}$ operating above its Gate 1 floor*
 220 *($\gamma > \gamma_{\text{min}}$) and Gate 2 floor ($\text{SNR} > \text{SNR}_{\text{min}}$), Gate 3 is the binding constraint whenever*
 221 *$\|\delta\boldsymbol{\theta}\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$.*

222 *Proof.* See Supplementary Note 1, Proposition 2. □

Key finding: Gate 3 dominates. Across all 14 validated configurations (12 modalities spanning 5 carrier families), **Gate 3** is the dominant failure gate ($C_{\text{mismatch}} > \max(C_{\text{noise}}, C_{\text{recover}})$) in 14/14 cases; Supplementary Tables S12–S13 confirm that Gates 1–2 bind only at extreme compression or photon starvation). In CASSI, a sub-pixel mask shift degrades MST-L by 13.98 ± 0.62 dB (mean $\pm$ s.d. over 10 scenes); in MRI, a 5% coil sensitivity mismatch under clinically realistic multi-coil conditions (8 coils, $4\times$ acceleration) degrades reconstruction by 1.75–7.14 dB (Supplementary Note 11); in cryo-EM, a 200 nm defocus error degrades Wiener-filter reconstruction by 1.1 dB; in ultrasound, a 200 m/s speed-of-sound error degrades DAS beamforming by 0.4 dB. Calibration—not solver design—is the underinvested bottleneck: a single forward-model correction recovers more reconstruction quality than the gap between traditional and state-of-the-art solvers.

Relationship to the Finite Primitive Basis. The two contributions are complementary: the Finite Primitive Basis provides a universal representation (every forward model is a DAG over 11 primitives¹), and the Triad provides a universal diagnostic law over that representation. The DAG structure makes Gate 3 diagnosis *actionable*: the MismatchAgent localizes the offending primitive node and corrects its parameters.

Consequences: Diagnosis and Correction

The two theoretical results directly imply a practical framework. The OPERATORGRAPH provides a modality-agnostic representation; the Triad provides root-cause diagnosis. Three deterministic agents—one per gate—evaluate information capacity, carrier budget, and operator mismatch respectively, producing a quantitative TRIADREPORT for every imaging configuration (see Methods for implementation details).

Correction pipeline. When **Gate 3** is dominant, a two-stage correction pipeline refines the forward model parameters: a coarse beam search over the mismatch parameter family $\theta = (\theta_1, \dots, \theta_k)$ associated with the offending DAG node, followed by gradient refinement via backpropagation through the differentiable forward model. Critically, correction operates exclusively on the forward model parameters, not on the solver weights: any existing solver—iterative, plug-and-play, or deep unrolling—benefits from the corrected operator without modification or retraining.

4-Scenario Protocol. To rigorously evaluate correction quality, PWM defines four canonical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator H_{true} . **Scenario II** (Mismatch): the solver uses the nominal operator H_{nom} on data generated by H_{true} . **Scenario III** (Corrected): the solver uses the PWM-corrected operator. **Scenario IV** (Oracle): the true operator on mismatched data, providing the correction ceiling.

The recovery ratio $\rho = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies how much of the mismatch-induced degradation is recovered (see Methods, Equation (5)).

Empirical Validation

We validate both theoretical contributions in two stages: controlled simulation experiments across twelve modalities using the 4-Scenario Protocol, and hardware validation on real data from ten modalities spanning all five carrier families—*incoherent photons* (CASSI, CACTI), *coherent photons* (compressive holography, fluorescence microscopy), *X-ray photons* (CT, CBCT), *electrons* (ptychography, cryo-EM), *nuclear spins* (MRI), and *acoustic waves* (ultrasound). This is, to our knowledge, the broadest cross-carrier validation of any computational imaging framework.

Simulation experiments

Correction results. Extended Data Table 1 (Supplementary Table S1) summarizes the oracle correction ceiling across 14 configurations spanning 12 modalities. The oracle gain $\Delta_{\text{oracle}} = \text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}$ ranges from $+0.76 \pm 0.12$ dB (CASSI/GAP-TV, 95% bootstrap CI over 10 scenes) to $+10.68 \pm 0.41$ dB (CT) across the original photon/X-ray modalities. Autonomous grid-search calibration achieves 82–100% of the oracle ceiling (Supplementary Table S9). The validated modalities span photon-domain (CASSI: $+0.76/+6.50$ dB [GAP-TV/MST-L]; CACTI: $+10.21 \pm 0.85$ dB; SPC: $+7.71/+10.38$ dB [FISTA-TV/HATNet]; Lensless: $+3.55 \pm 0.22$ dB; compressive holography: $+0.0/+0.4$ dB; fluorescence microscopy: $+0.1/+6.8$ dB), coherent-photon (Ptychography: $+7.09 \pm 0.47$ dB), X-ray (CT: $+10.68 \pm 0.41$ dB; CT offset: $+2.5/+3.9$ dB), acoustic (Ultrasound: $+0.1/+0.9$ dB), and electron (Cryo-EM: $+0.1/+3.6$ dB) and nuclear spin (MRI: $+1.75$ to $+7.14$ dB under clinically realistic multi-coil conditions at 3–5% sensitivity mismatch; Supplementary Note 11). All confidence intervals are computed via the bootstrap percentile method with $B = 1,000$ resamples over test scenes (Methods). Correction gains are commensurate across all five carrier families, confirming carrier-agnostic operation.

Modality deep dives. In CASSI, a 5-parameter mismatch¹⁷ collapses all solvers to ~ 21 dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ($\rho_{\text{IV}} = 0-0.57$; Supplementary Table S2), confirming that multi-parameter mismatches are harder to correct than isolated shifts. In CACTI, EfficientSCI¹⁸ drops by 20.58 dB, yet GAP-TV achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S9)—the best ideal-condition solver suffers the largest degradation. SPC gain drift is corrected to 86–92% of the oracle ceiling (Table S9). **Gate 3** is dominant in every case; Gates 1–2 impose irrecoverable information-theoretic limits only at extreme compression or photon starvation (Supplementary Tables S12–S13).

Zero-shot generalization. Hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains, with comparable correction gains (Figure 4), confirming carrier-agnostic operation enabled by the OperatorGraph abstraction.

Hardware validation

CASSI real data. These experiments use physical measurements from hardware-calibrated coded aperture systems—not synthetic data—providing direct evidence that mismatch dominance persists under real manufacturing tolerances and environmental conditions. We reconstruct 5 real TSA scenes¹⁰ under calibrated and perturbed masks. GAP-TV shows a mean residual ratio of $1.8\times$ ($0.00189 \rightarrow 0.00333$); HDNet shows $1.0\times$ (mask-oblivious). MST-S/L show ratios near $1.0\times$ on real data, in contrast to severe synthetic degradation. This asymmetry is itself informative: real hardware already contains unmodeled manufacturing imperfections, and an additional software-simulated perturbation is partially absorbed by pre-existing calibration errors (Supplementary Table S7). An extended cross-residual analysis (Supplementary Note 15) reveals a deeper diagnostic: the *cross-residual*—evaluating a mismatched reconstruction under the true forward model—increases monotonically from 0.3% at 0.25 px shift to 11.1% at 2.0 px shift, with per-scene standard deviation $< 0.4\%$, confirming that Gate 3 sensitivity is scene-independent and monotonic in mismatch magnitude. These real-data results independently confirm the Triad predictions from simulation: the InverseNet benchmark¹⁷ validates the same mismatch-dominance pattern on physical DD-CASSI and CACTI instruments.

CACTI real data. GAP-TV shows $10.4\times$ mean residual ratio under sub-pixel shift, with per-scene ratios from $9.4\times$ to $11.0\times$. An extended analysis (Supplementary Note 15) reveals a striking dissociation: the *self-residual* barely changes ($1.0\text{--}1.12\times$) because the solver adapts to the wrong model, but the *cross-residual* explodes $44\text{--}462\times$ as the shift increases from 0.25 to 1.0 px. This demonstrates that self-consistency is not a reliable diagnostic for operator mismatch in underdetermined systems, with direct implications for compressive sensing quality control.

CT real sinograms. We extend hardware validation beyond optical instruments to three additional carrier families. For **X-ray CT**, two public sinogram datasets—the FIPS walnut micro-CT (1200 projections, 2296 detectors) and the Helsinki Tomography Challenge 2022 (721 projections, 560 detectors)—show that center-of-rotation (CoR) offsets of 1–8 pixels cause monotonic PSNR degradation of 9.4 dB (walnut) and 8.0 dB (HTC), with 100% oracle recovery (Supplementary Note 15). For **electron ptychography**, real 4D-STEM data from SrTiO₃ (Zenodo 5113449; 300 kV, 128×128 scan) shows that probe position jitter of 0.5–4.0 pixels causes monotonic iCoM phase degradation from 35.5 to 19.4 dB (16.1 dB total loss),

with $> 99.9\%$ oracle recovery (Supplementary Note 15). For **MRI**, multi-coil brain k -space from M4Raw (Zenodo 8056074; 4 coils, 256×256) shows that coil sensitivity perturbation up to 40% degrades R=2 SENSE reconstruction by 0.5 dB, with 100% recovery via sensitivity re-estimation. The modest MRI effect is consistent with the over-determined encoding at R=2; the simulation experiments at R=4 (Supplementary Note 11) confirm that Gate 3 severity scales with acceleration factor.

Ultrasound. The acoustic carrier family—absent from all prior validation—is tested via simulated pulse-echo B-mode imaging with a 64-element linear array at 5 MHz centre frequency and 100 MHz sampling rate on a 128×128 tissue phantom. Speed-of-sound (SoS) mismatch of $[50, 100, 200, 400]$ m/s (relative to true $c = 1540$ m/s) causes monotonic PSNR degradation from 0.1 to 0.9 dB, with delay-and-sum (DAS) beamforming as the reconstruction solver. Grid-search calibration over 21 SoS candidates in $[1440, 1640]$ m/s achieves $\rho = 1.19$ – 2.77 (Supplementary Table S14); the over-unity ratios indicate that the optimizer consistently finds an operating point ($c = 1440$ m/s) that improves upon oracle reconstruction, reflecting the DAS beamformer’s inherent model simplifications. This confirms Gate 3 dominance for acoustic propagation, completing the five-carrier-family validation.

Cryo-EM. Electron-carrier validation uses simulated cryo-EM micrographs (128×128 , pixel size 0.1 nm) with contrast transfer function (CTF) encoding. Defocus mismatch of $[100, 200, 500, 1000]$ nm (relative to true $\Delta f = -2000$ nm) degrades Wiener-filter reconstruction by 0.1–3.6 dB. At Nyquist frequency 5 nm^{-1} , the CTF exhibits ~ 30 oscillations before the B-factor envelope ($B = 2 \text{ nm}^2$) attenuates high frequencies, making the reconstruction highly sensitive to defocus error. Grid-search calibration over 51 defocus values in $[-3000, -500]$ nm recovers $\rho = 1.00$ – 1.12 (Supplementary Table S15), confirming that CTF parameter mismatch is the dominant bottleneck in single-particle reconstruction—consistent with the extensive defocus-estimation literature in structural biology.

CT detector offset. X-ray CT validation uses a 128×128 Shepp–Logan phantom with 360-angle parallel-beam projections. Detector offset mismatch of $[2, 5, 10, 20]$ pixels—simulated by shifting the sinogram along the detector axis—causes 2.5–3.9 dB PSNR degradation under filtered backprojection (FBP) with Shepp–Logan filter. At large offsets (≥ 10 px), edge data loss additionally reduces the oracle PSNR from 13.8 to 10.6 dB (a Gate 1 effect). Grid-search calibration over 21 offset candidates in $[-25, 25]$ px achieves $\rho = 0.83$ – 1.00 (Supplementary Table S16), with exact offset recovery (≤ 0.5 px error) for offsets ≥ 5 pixels.

Compressive holography. Multi-depth Fresnel propagation encodes a 3D object ($4 \times 64 \times 64$) into a single 2D hologram—the most compressive encoding among validated modalities ($\lambda = 532$ nm, pixel pitch $5 \mu\text{m}$, depth spacing $200 \mu\text{m}$). Propagation distance error of $[10, 50, 100, 200]$ μm causes progressive defocus, degrading FISTA-TV reconstruction by up

364 to 0.4 dB at 200 μm error. Autonomous calibration via hologram residual minimisation
 365 ($\|y - A_{\text{cal}}\hat{x}\|^2/\|y\|^2$) achieves $\rho = 0.64\text{--}1.59$ (Supplementary Table S17). The per-plane
 366 PSNR analysis reveals that deeper planes are more sensitive to distance error, consistent
 367 with the quadratic phase scaling of Fresnel kernels.

368 **Fluorescence microscopy.** Dual-PSF Stokes-shift imaging—excitation and emission wave-
 369 lengths produce distinct PSF widths—is validated under PSF sigma mismatch of $[0.1, 0.3, 0.5, 1.0]$ pixels
 370 on a 64×64 fluorescence phantom (1000 peak photons). Richardson–Lucy deconvolution
 371 (80 iterations) degrades by 0.1–6.8 dB. A 2D grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ with measurement-
 372 residual minimisation achieves $\rho = 0.44\text{--}0.75$ (Supplementary Table S18). Recovery is
 373 lowest at the smallest error ($\Delta\sigma = 0.1$ px, $\rho = 0.44$) where the calibration signal is near the
 374 noise floor, and strongest at moderate errors ($\Delta\sigma = 0.5$ px, $\rho = 0.75$) where the objective
 375 landscape is well-resolved.

376 **Autonomous calibration.** Grid-search calibration recovers $\rho = 0.85$ (CASSI, 1,140 s;
 377 95% CI $[0.79, 0.91]$), $\rho = 1.00$ (CACTI, 60 s; CI $[0.97, 1.00]$), $\rho = 0.86\text{--}0.92$ (SPC via TV
 378 objective; CI width ≤ 0.06), $\rho = 1.19\text{--}2.77$ (Ultrasound SoS), $\rho = 1.00\text{--}1.12$ (Cryo-EM
 379 defocus), $\rho = 0.83\text{--}1.00$ (CT detector offset), $\rho = 0.64\text{--}1.59$ (compressive holography),
 380 and $\rho = 0.44\text{--}0.75$ (fluorescence PSF) of the oracle correction. Recovery ratios exceeding
 381 unity (ultrasound, cryo-EM, compressive holography) indicate that the optimizer found
 382 operating points marginally superior to the assumed ground-truth parameters, reflecting
 383 inherent model simplifications in the reconstruction algorithms. Single-parameter modal-
 384 ities (CACTI, CT) achieve near-perfect or above-unity recovery because their mismatch
 385 manifolds are low-dimensional with clean objective landscapes. CT CoR calibration on real
 386 sinograms also achieves $\rho = 1.00$ (CI $[0.99, 1.00]$). Multi-parameter modalities (CASSI, flu-
 387 orescence) show lower but still substantial recovery, reflecting the higher-dimensional search
 388 space.

389 **Simulation-to-hardware gap.** CASSI shows smaller real degradation ($1.8\times$) than pre-
 390 dicted by simulation, because as-built masks already contain unmodeled imperfections that
 391 absorb incremental perturbations. CACTI shows larger real sensitivity ($10.4\times$) than simula-
 392 tion, because the temporal mask pattern replicates errors across all compressed frames. This
 393 bidirectional discrepancy—invisible without a unified cross-modality framework—carries a
 394 methodological lesson: synthetic mismatch studies can both overestimate marginal impact
 395 (CASSI) and underestimate cumulative sensitivity (CACTI).

396 Discussion

397 The central finding is that the space of imaging forward models has finite structural
 398 complexity—11 primitives suffice and are necessary—and that operator mismatch, not in-

formation deficiency or noise, is the dominant reconstruction bottleneck across all validated modalities and all five carrier families. The twelve validated modalities span the breadth of modern imaging science: from coded aperture cameras (CASSI, CACTI) through clinical scanners (CT, CBCT, MRI, ultrasound) to instruments at the frontier of structural biology (cryo-EM, 2017 Nobel Prize in Chemistry) and super-resolution microscopy (fluorescence, 2014 Nobel Prize in Chemistry). In every case, the same 11 primitives and 3 gates apply. The implication is not that solver research is unnecessary, but that calibration is systematically underinvested: a single forward-model correction step often recovers more reconstruction quality than the gap between a classical solver and a state-of-the-art deep network operating on the same mismatched operator.

Implications for autonomous scientific discovery. The bounded complexity of imaging physics has consequences that extend beyond calibration. If every imaging forward model—from a \$10 webcam to a particle accelerator beamline—decomposes into the same 11 primitives, then any autonomous system tasked with “learning to see” faces a *finite search problem* over typed physical operators, not an open-ended function approximation. This re-frames the challenge for AI Scientists¹⁹: rather than discovering physical laws from scratch, a machine learning system equipped with the OPERATORGRAPH representation needs only to identify which subset of 11 known primitives is active in a new instrument and estimate their parameters from data. The Triad Decomposition further constrains the hypothesis space for failure: an AI Scientist confronting a degraded reconstruction has exactly three root causes to consider, each with a formal test and a known correction strategy. The observation that 9 of 11 primitives are introduced by the first 10 modalities, with the final two (Disperse and Scatter) appearing only for exotic carrier physics, suggests that the primitive manifold is dense at its core and sparse at its boundary—a structure that should enable few-shot generalization to new modalities from existing primitive combinations. More broadly, the Finite Primitive Basis Theorem raises a question for computational science beyond imaging: if the forward model of every physical measurement system can be expressed in a small universal grammar, what analogous grammars exist for other physical processes, and what does their existence imply for the long-term trajectory of physics-informed machine learning?

Implications for biology and medicine. The framework has immediate relevance for the two largest imaging user communities. In structural biology, cryo-EM resolution is rate-limited by contrast transfer function (CTF) estimation—a textbook Gate 3 problem. A 200 nm defocus error at 300 kV limits the achievable resolution from $\sim 2 \text{ \AA}$ to $\sim 3 \text{ \AA}$, and CTFFIND-class estimators require thousands of particles to converge. The PWM Triad diagnoses this as Gate 3 dominance with a 1-parameter correction manifold (Δf), and autonomous defocus calibration recovers $\rho = 1.00\text{--}1.12$ of the oracle (Supplementary Table S15). In clinical imaging, CBCT scatter and beam hardening limit image quality

in radiation oncology treatment planning; ultrasound speed-of-sound inhomogeneity causes geometric distortion and resolution loss in abdominal imaging; and MRI coil sensitivity drift degrades parallel imaging reconstruction. All three failure modes are instances of Gate 3, and all three are correctable through the same forward-model calibration pipeline. The unification of these disparate calibration challenges under a single diagnostic framework suggests a path toward automated quality assurance across the clinical imaging fleet.

Comparison with specialist calibration. Specialist calibration methods—ESPIRiT⁸ for MRI, ePIE⁹ for ptychography, cross-correlation auto-focus for CT, CTFFIND for cryo-EM—outperform PWM on their home modality when adequate calibration data is available (Supplementary Note S14). However, they degrade under data-limited conditions (ESPIRiT: -9.29 dB at 24 ACS lines), require per-modality engineering, and do not exist for several validated modalities (CASSI, CACTI, compressive holography, ultrasound SoS correction). PWM’s value is *generality*: a single pipeline serving 12 modalities across all five carrier families—including two Nobel Prize-winning techniques and four clinical instruments—with robust performance across calibration data regimes. The two approaches are complementary—specialist estimates can initialize PWM correction, and PWM can refine specialist estimates.

Hardware validation interpretation. The hardware results reveal that Gate 3 sensitivity depends on the *condition number* of the encoding matrix. On real CASSI, perturbation produces smaller degradation ($1.8\times$) than predicted, because the as-built mask already contains unmodeled manufacturing imperfections. On real CACTI, degradation is severe ($10.4\times$), because the simpler temporal-mask optics lack this pre-existing error buffer. On real CT sinograms, CoR offset causes 8–9 dB loss; electron ptychography shows up to 16.1 dB phase degradation; ultrasound SoS mismatch causes 0.1–0.9 dB loss; cryo-EM defocus error causes up to 3.6 dB degradation; and CBCCT detector offset produces 2.5–3.9 dB loss. This range—from sub-dB (well-conditioned MRI at $R=2$, ultrasound) to > 16 dB (ptychography)—is itself a Triad prediction: the condition number of the encoding matrix determines mismatch sensitivity, with well-conditioned systems showing graceful degradation and ill-conditioned systems showing catastrophic failure. The extended cross-residual analysis further reveals that *self-consistency metrics are misleading* in underdetermined systems (CACTI cross-residual is $462\times$ while self-residual is only $1.12\times$), underscoring the need for forward-model-aware diagnostics.

Boundary conditions and failure modes. The framework has well-defined limitations that inform its scope of applicability:

- **Multi-parameter mismatch:** CASSI recovery ratios under 5-parameter mismatch are moderate ($\rho = 22\text{--}46\%$), reflecting the inherent difficulty of simultaneously cor-

recting multiple coupled parameters on a high-dimensional manifold. Single-parameter correction achieves 85–100%.

- **Nonlinear forward models:** The Triad diagnostic has been validated on linear forward models only; beam hardening CT and phase-wrapped MRI require additional validation (the 11-primitive basis formally covers these via Transform Λ^1).
- **Undeclared mismatch:** Correction is limited to declared mismatch parameter families in the mismatch database; undeclared failure modes (e.g., nonlinear detector drift) require extending the database.
- **Software-perturbation protocol:** Current real-data experiments apply software-simulated perturbations to existing measurements rather than physically displacing hardware and re-acquiring; the controlled hardware protocol is specified in Methods and Supplementary Note 8.

Falsifiable predictions. The framework generates concrete, testable predictions:

1. **SIM:** Gate 3 should dominate when pattern phase error exceeds 0.05 rad. Predicted correction: +3–8 dB.
2. **OCT:** Dispersion mismatch should produce +2–5 dB correction gain via the Disperse primitive (W).
3. **Electron ptychography:** Position errors exceeding 1/10 of the probe diameter should trigger Gate 3 dominance. This prediction is *confirmed* by real 4D-STEM validation: +5 to +16 dB correction on SrTiO₃ data (Supplementary Note 15).
4. **Cryo-EM:** CTF defocus mismatch of > 200 nm should shift the FSC_{0.143} resolution threshold by > 0.5 Å. Our simulation confirms 1.1–3.6 dB degradation at 200–1000 nm defocus error.
5. **Ultrasound:** Speed-of-sound mismatch of > 100 m/s should produce > 1 mm geometric distortion in B-mode images at clinical imaging depths. Our DAS simulation confirms monotonic degradation of 0.1–0.9 dB.
6. **CBCT:** Geometric calibration drift of > 5 pixels should produce cupping artifacts with > 3 dB degradation. Our FBP simulation confirms 3.9 dB loss at 5-pixel detector offset with 100% oracle recovery.

These predictions are falsifiable: if any modality shows Gate 1 or Gate 2 dominance under the stated conditions, the framework would require revision. Three of six predictions are now confirmed by simulation or hardware data. Hardware-in-the-loop validation with physical parameter displacement is the natural next step (Supplementary Note 8).

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510 TION framework, proved the Finite Primitive Basis Theorem, developed the OPERATOR-
511 GRAPH IR, implemented the agent and correction systems, performed all simulation and
512 real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruc-
513 tion algorithm used as the primary solver across CASSI, CACTI, and SPC experiments,
514 contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI
515 and CACTI forward model specifications and mismatch parameter characterizations that
516 define the 5-parameter mismatch model, validated the real-data experimental protocols for
517 both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

518 **Competing Interests.** C.Y. is an employee of NextGen PlatformAI C Corp, which de-
519 velops the PWM platform. The authors declare no other competing interests.

520 **Ethics Declarations.** This study does not involve human participants, human tissue, or
521 animals. All experiments use publicly available benchmark datasets and simulated data.

522 **Data Availability.** All synthetic measurement data can be regenerated using the OPER-
523 ATORGRAPH templates and mismatch parameters in the Supplementary Information. The
524 KAIST hyperspectral dataset²⁰ and TSA real-data scenes used for CASSI experiments are
525 publicly available. CACTI real-data scenes are available from the EfficientSCI repository¹⁸.
526 CT sinograms are from the FIPS walnut dataset (Zenodo 1254206) and Helsinki Tomogra-
527 phy Challenge 2022 (Zenodo 6984868). The 4D-STEM ptychography dataset (SrTiO₃ [001])
528 is from Zenodo 5113449. Multi-coil MRI k -space data (M4Raw) is from Zenodo 8056074.
529 Ultrasound, cryo-EM, CBCT, compressive holography, and fluorescence microscopy exper-
530 iments use procedurally generated phantoms with parameters and random seeds fully spec-
531 ified in Methods; all scripts are included in the repository.

532 **Code Availability.** The PWM codebase, including all OPERATORGRAPH templates, agent
533 implementations, real-data validation scripts, and evaluation pipelines, is available at https://github.com/integritynoble/Physics_World_Model
534 under the PWM Noncommercial
535 Share-Alike License v1.0 (see LICENSE in the repository).

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Online Methods

OperatorGraph Specification

Formal definition. The OPERATORGRAPH intermediate representation encodes the forward physics of any computational imaging modality as a directed acyclic graph (DAG) $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points: `forward( $x$ )  $\rightarrow$   $y$` and `adjoint( $y$ )  $\rightarrow$   $x$` , the latter defined only when the primitive is linear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each node additionally exposes a set of learnable parameters θ_i that may be perturbed during mismatch simulation or optimized during calibration, as well as read-only metadata flags (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator` object by the `GraphCompiler`.

Node types. Primitive operators fall into two categories:

- **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), subpixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encoding (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation (`fresnel_propagate`), random projection (`random_project`), and structured illumination (`sim_modulate`). Each implements both `forward()` and `adjoint()`.
- **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlinearity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` = `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–Saxton-type algorithms).

Adjoint consistency and Lipschitz bounds as mathematical safety rails. Two structural constraints distinguish the OPERATORGRAPH IR from standard deep-learning forward models and prevent either nonlinear primitive family from becoming a universal approximator.

First, *adjoint consistency*: correctness of every linear primitive is verified by a randomized dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^* y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^* y, x \rangle|, \epsilon)} \quad (1)$$

where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-6}$. At the graph level,

a compiled `GraphOperator` composed entirely of linear nodes executes the same test over the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} , and $\bar{\delta}$ for audit. All graph templates that consist solely of linear primitives pass this check. Adjoint consistency is not merely a numerical correctness guarantee: it is a *physical faithfulness certificate*. A model whose adjoint is inconsistent implements a physically impossible energy accounting, making gradient-based calibration unreliable and certifiable reconstruction bounds vacuous. Neural network forward models lack this certificate by construction, since their weight matrices have no physical meaning and their transposes are not computed or validated.

Second, *Lipschitz bounds*: the two nonlinear primitive families, Detect D and Transform Λ , are each restricted to five canonical function families with at most two scalar parameters. This restriction enforces bounded Lipschitz constants ($\text{Lip}(\Lambda_k) \leq L$ for the explicit L derived in¹) and prevents either primitive from becoming a universal approximator in the sense of Cybenko²¹. Concretely: a generic neural network layer with sigmoid activations can approximate any continuous function on a compact domain; neither Detect nor Transform can, because their function families are closed under composition only within the five enumerated types. This non-universality is the key property that makes the 11-primitive library *falsifiable*: if a modality’s forward physics cannot be represented within the restricted families, the extension protocol (Main Text, “Extension protocol”) fires and a new primitive must be justified physically. The combination of adjoint consistency and bounded Lipschitz nonlinearity provides the mathematical safety rails that make OPERATORGRAPH-based reconstruction both interpretable and certifiable—properties that standard physics-informed neural networks²² achieve only approximately and without modality-agnostic guarantees.

Graph compilation. The compiler executes a four-stage pipeline:

1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that every `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id` values, and optionally verify shape compatibility along edges when a `canonical_chain` metadata flag is set.
2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|V|)})$.
4. **Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the topological order and applies each node’s individual adjoint in sequence, implementing the chain rule $A^* = A_1^* \circ \dots \circ A_{|V|}^*$ for a composition $A = A_{|V|} \circ \dots \circ A_1$. For graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to `adjoint()` raises `NotImplementedError` at runtime.

The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for provenance tracking in RunBundle manifests.

Template library. The `graph_templates.yaml` registry contains 90 templates spanning all five carrier families. Of these, 26 modalities carry formal validation status (7 with full end-to-end correction validation, 1 with Scenario I baseline, 18 with template-level validation; Supplementary Table S3). Representative modalities by carrier family include:

- **Photons (optical and X-ray):** CASSI, SPC, CACTI, structured illumination microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptychographic microscopy (FPM), optical coherence tomography (OCT), lensless imaging, light field, integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluorescence lifetime imaging (FLIM), diffuse optical tomography (DOT), phase retrieval, X-ray computed tomography (CT), and cone-beam CT (CBCT).
- **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron energy loss spectroscopy (EELS), and electron holography.
- **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and magnetic resonance spectroscopy (MRS).
- **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar, and photoacoustic tomography (combines optical excitation with acoustic detection).
- **Particles:** Neutron tomography, proton radiography, and muon tomography.

Fidelity levels. Each template is parameterized by a fidelity level that controls the degree of physical realism in the simulated forward model:

Level 1 (Linear, shift-invariant): The forward model is a linear, spatially uniform operator—the simplest approximation, suitable for initial diagnostics and rapid prototyping.

Level 2 (Linear, shift-variant): Spatially varying operator parameters (e.g. non-uniform illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for photon-counting modalities, Rician noise for MRI, Poisson for CT).

Level 3 (Nonlinear, ray/wave-based): Includes nonlinear effects such as beam hardening, phase wrapping, and scattering, captured by the Transform Λ primitive. Perturbation families and ranges are specified in `mismatch.db.yaml`.

Level 4 (Full-wave / Monte Carlo): Complete physical simulation including wave-optical propagation, spatially varying aberrations, detector nonlinearities, and environmental drift. Currently implemented for holography and ptychography.

All 11 primitives in the library $\mathcal{B}$ apply uniformly across every fidelity level; the levels organize simulation complexity, not theorem scope.

Triad Decomposition Formalization

The TRIAD DECOMPOSITION asserts that the quality of any computational imaging reconstruction is bounded by three fundamental gates. Rather than a qualitative guideline, PWM quantifies each gate numerically and uses the resulting scores to diagnose the dominant bottleneck in any imaging configuration.

Gate 1 (Recoverability). Recoverability measures the information-theoretic capacity of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where m is the number of independent measurements and n the dimension of the signal. The `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table mapping compression ratio to expected reconstruction PSNR under ideal conditions, obtained from calibration experiments or published benchmarks. Each entry carries a **provenance** field citing the source (paper DOI, internal experiment ID, or theoretical formula). Additional recoverability indicators include the effective rank of the measurement matrix (estimated via randomized SVD for large operators), the dimension of the null space, and the restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian random projections in SPC).

Gate 2 (Carrier Budget). The carrier budget quantifies the signal-to-noise ratio (SNR) of the measurement channel. The **PhotonAgent** consumes the `photon_db.yaml` registry (624 lines) which stores, per modality, a deterministic photon model parameterized by source power, quantum efficiency, exposure time, and detector characteristics. The agent classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated, $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$), and *dark-current-limited* (long exposures where dark current accumulation dominates). The output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**, or **insufficient**).

Gate 3 (Operator Mismatch). Operator mismatch quantifies the discrepancy between the assumed forward model H_{nom} and the true physical operator H_{true} . The **MismatchAgent** consults `mismatch_db.yaml` (797 lines) which catalogs, for each modality, the set of mismatch parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available correction methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalized ℓ_2 distance $\|\theta_{\text{true}} - \theta_{\text{nom}}\|/\|\theta_{\text{range}}\|$, where θ_{range} is the per-parameter dynamic range from the registry. Sensitivity analysis $\partial \text{PSNR}/\partial \theta_k$ is estimated via finite differences on the forward model. The output is a **MismatchReport** containing the severity score, the dominant mismatch parameter, the recommended correction method, and the expected PSNR gain from

677 correction.

678 **Gate binding determination.** Given reconstruction results under the four-scenario pro-
679 tocol (the Evaluation Protocol section below), PWM identifies the dominant gate by com-
680 paring three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

681 where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under
682 Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition,
683 and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant
684 gate is $\arg \max_g C_g$.

685 **TriadReport.** For every diagnosis, PWM produces a TRIADREPORT: a Pydantic-validated
686 structured artifact comprising `dominant_gate` (enum: `recoverability`, `carrier_budget`,
687 `operator_mismatch`), `evidence_scores` (three floats, one per gate), `confidence_interval`
688 (float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compres-
689 sion ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary
690 mapping each mismatch parameter name to its $\partial \text{PSNR} / \partial \theta_k$ value). The TRIADREPORT is
691 mandatory—PWM does not permit a reconstruction to be reported without an accompany-
692 ing diagnosis.

693 **Recovery ratio.** We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

694 which lies in $[0, 1]$ under standard convexity conditions (see Supplementary Note 1 for
695 formal analysis; values $\rho > 1$ are possible when the corrected operator provides beneficial
696 regularization). $\rho = 0$ indicates that calibration yields no benefit (mismatch is not the
697 bottleneck), while $\rho = 1$ indicates that calibration fully closes the mismatch gap.

698 Agent System Architecture

699 The PWM agent system comprises 6 specialist agents, 1 optional hybrid agent, and 8
700 support classes totalling 10,545 lines of Python. All agents execute deterministically; no
701 large language model (LLM) is required for pipeline operation.

702 **PlanAgent.** The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`,
703 PlanAgent parses the intent (`simulate`, `operator_correction`, or `auto`), maps the re-

704 requested modality to its canonical key via the `modalities.yaml` registry (which contains 64
 705 modality entries with keywords, forward model equations, and default solvers), builds an
 706 `ImagingSystem` contract, and dispatches to the appropriate sub-agents. When the mode is
 707 `auto`, `PlanAgent` inspects the available data and operator specification to determine whether
 708 simulation or operator correction is more appropriate.

709 **PhotonAgent.** Computes SNR feasibility deterministically from the `photon_db.yaml`
 710 registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the agent
 711 evaluates the photon budget by combining source power, quantum efficiency, exposure time,
 712 and noise model parameters. The output `PhotonReport` is a strict Pydantic model contain-
 713 ing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and `per_element_photons`
 714 (float).

715 **RecoverabilityAgent.** A table-driven agent that consults `compression_db.yaml` (1,186
 716 lines) to map the modality and compression ratio to an expected PSNR range. Each table
 717 entry includes provenance metadata citing the original source. The output `RecoverabilityReport`
 718 contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where
 719 available.

720 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration us-
 721 ing `mismatch_db.yaml` (797 lines). For each modality, the database enumerates the rel-
 722 evant mismatch parameters, their physical units, typical perturbation ranges, and avail-
 723 able correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1),
 724 `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).

725 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recover-
 726 ability, and Mismatch agents, computes the gate costs (Equations (2) to (4)), identifies the
 727 dominant gate, and generates actionable suggestions. The `AnalysisAgent` also computes
 728 the recovery ratio ρ and its bootstrap confidence interval.

729 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is au-
 730 thorized, the negotiator inspects all three upstream reports and applies three veto con-
 731 ditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover}
 732 both large); (2) severe mismatch (severity > 0.7) without a planned correction step; (3) joint
 733 probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems
 734 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable
 735 explanation and suggested remediation.

736 **HybridAgent.** An optional wrapper that invokes an LLM for natural-language narra-
 737 tive generation or edge-case modality mapping. All quantitative decisions remain on the

738 deterministic code path; the HybridAgent is never required for pipeline operation.

739 **Support classes.** The remaining components include: **AssetManager** (file I/O and caching
740 for large arrays), **ContinuityChecker** (verifies that sequential pipeline outputs are dimen-
741 sionally consistent), **SystemDiscern** (auto-detects modality from uploaded data), **PreflightChecker**
742 (validates the complete experiment configuration before execution), **WhatIfPrecomputer**
743 (evaluates counterfactual what-if scenarios), **SelfImprovement** (logs diagnostic events for
744 future registry refinement), **PhysicsStageVisualizer** (generates intermediate visualiza-
745 tions at each pipeline stage), and **UPWMI** (Universal Physics World Model Interface, the
746 top-level entry point that wires all agents together).

747 **Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All
748 contracts inherit from **StrictBaseModel**, which enforces `extra="forbid"` (no unexpected
749 fields), `validate_assignment=True` (mutations re-validated), and a model validator that
750 rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums
751 are string enums for human-readable JSON serialization. This design ensures that pipeline
752 failures surface immediately as validation errors rather than propagating silently.

753 **YAML registries.** The system is driven by 9 YAML registries totalling 7,034 lines:
754 `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skele-
755 tons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and
756 correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml`
757 (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml`
758 (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds
759 per metric).

760 Correction Algorithms

761 We implement two complementary algorithms for operator mismatch correction. Crucially,
762 both algorithms operate on the forward operator parameters θ rather than the reconstruc-
763 tion solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any
764 downstream solver (GAP-TV, MST-L, HDNet²³, CST, *etc.*) without retraining.

765 **Algorithm 1: Hierarchical Beam Search.** The coarse correction phase employs a
766 hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI,
767 the five-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy
768 and rotation θ) and dispersion parameters (slope a_1 and axis angle α); an optional sixth
769 parameter, PSF width σ_{psf} , is available but not used in the primary experiments. The
770 algorithm proceeds as follows:

- 771 1. **1D sweeps.** Each parameter is swept independently over its full range while holding
772 others at nominal values. This produces five 1D cost curves from which coarse optima
773 are extracted.
- 774 2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$
775 grid centered on the 1D optima. The top- k ($k = 5$) candidates by reconstruction
776 PSNR are retained.
- 777 3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, α)
778 is searched over a 5×7 grid. The joint top- k candidates are retained.
- 779 4. **Coordinate descent refinement.** Three rounds of univariate refinement on each
780 parameter, shrinking the search interval by factor 2 at each round, produce the final
781 estimate $\hat{\theta}_{\text{Alg1}}$.

782 Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is
783 ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

784 **Algorithm 2: Joint Gradient Refinement.** The fine correction phase uses a differen-
785 tiable forward model to jointly optimize all mismatch parameters via gradient descent. The
786 key components are:

- 787 1. **Differentiable mask warp.** The binary mask is warped by a continuous affine
788 transformation using bilinear interpolation, implemented as a custom PyTorch module
789 (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-
790 through estimator (STE) to maintain binary structure while permitting gradient flow.
- 791 2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is
792 implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`)
793 that accepts mismatch parameters as differentiable inputs.
- 794 3. **GPU grid initialization.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 =$
795 567 points provides diverse starting candidates. The top 9 candidates seed multi-start
796 gradient refinement.
- 797 4. **Staged gradient refinement.** Each of the 9 candidates is refined using Adam
798 optimization (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
799 4 random restarts with jittered initialization guard against local minima. The loss
800 function is the negative PSNR computed via an unrolled K -iteration differentiable
801 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

802 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
803 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3–5 $\times$ improve-
804 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
805 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

Evaluation Protocol

Four-Scenario Protocol. We evaluate every modality under four standardized scenarios that isolate different sources of quality degradation:

Scenario I (Ideal): $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . In this scenario the system is perfectly calibrated ($H_{\text{true}} = H_{\text{nom}}$), so the operator used for reconstruction matches the one that generated the data. This yields the oracle upper bound on reconstruction quality, limited only by the sensing geometry and solver convergence.

Scenario II (Mismatch): $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq H_{\text{true}}$). This is the standard operating condition in practice: the measurement is generated by the true physics, but the reconstruction uses a nominal (potentially mismatched) forward model.

Scenario III (Corrected): $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\theta})$ where $\hat{\theta}$ is estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

Scenario IV (Oracle Mask): Same measurements as Scenario II ($\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ with $H_{\text{true}} \neq H_{\text{nom}}$); reconstruct with H_{true} instead of H_{nom} . Provides the correction ceiling: the best reconstruction achievable when the true operator is known exactly, applied to data that were sensed by the mismatched system. The gap between Scenario IV and Scenario I reveals the irreducible loss from the degraded sensing configuration itself (e.g., a shifted mask pattern is suboptimal even when perfectly known).

Metrics. Reconstruction quality is assessed using three complementary metrics:

- **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene and averaged. For signals normalized to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC data normalized to $[0, 255]$, the peak value is 255.
- **SSIM** (structural similarity index): captures perceptual quality including luminance, contrast, and structural components, computed with a Gaussian window of width 11 and standard deviation 1.5.
- **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the angle between predicted and true spectral vectors at each spatial location, reported in degrees. Lower is better.

Datasets.

- **CASSI:** 10 scenes from the KAIST dataset²⁰, each a $256 \times 256 \times 28$ spectral cube (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.

838 • **CACTI**: 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per
839 snapshot). Data range $[0, 1]$.

840 • **SPC**: 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data
841 range $[0, 255]$.

842 All per-scene metrics are reported individually as well as averaged, and all reconstruction
843 arrays are saved as NumPy NPZ files.

844 Experimental Details

845 **Hardware.** All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam
846 search) and all solver-based reconstructions use the GPU for matrix–vector products and
847 FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic
848 differentiation on the same GPU.

849 **CASSI configuration.** The coded aperture snapshot spectral imaging (CASSI) system
850 uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along
851 the spatial dimension. The five-parameter mismatch model $\theta = (dx, dy, \theta, a_1, \alpha)$ describes:
852 mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle
853 (θ , degrees), dispersion slope (a_1 , pixels per band), and dispersion axis angle (α , degrees).
854 An optional sixth parameter, PSF blur width (σ_{psf} , pixels), is available but not used in the
855 primary experiments. These mismatch parameter values were determined through system-
856 atic characterization of realistic CASSI assembly errors (Supplementary Note 8). The true
857 mismatch parameters are $\theta_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha =$
858 $0.15^\circ)$. Solvers evaluated include TwIST²⁴, GAP-TV²⁵, DGSMP²⁶, MST-L⁶, and CST-
859 L²⁷, all of which receive the same operator and differ only in their reconstruction algorithm.
860 The supplementary per-scene analysis additionally includes DeSCI²⁸ and HDNet²³.

861 **CACTI configuration.** The coded aperture compressive temporal imaging system uses
862 binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single
863 snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-
864 frame shift). The default solver is EfficientSCI¹⁸.

865 **SPC configuration.** The single-pixel camera uses random binary measurement patterns
866 at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch
867 is modeled as an exponential gain drift ($g_i = \exp(-\alpha \cdot i)$) on the measurement matrix. The
868 default solver is FISTA-TV with total-variation regularization.

869 **MRI configuration.** Cartesian k -space sampling with $4\times$ acceleration (25% of k -space
870 lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity

871 maps used for parallel imaging reconstruction. The default solver is SENSE¹⁵ with ℓ_1 -
872 wavelet regularization.

873 **ESPIRiT comparison protocol.** To quantitatively compare PWM with ESPIRiT⁸, we
874 evaluate four conditions on the same multi-coil MRI configuration: (1) Scenario I (true
875 maps), (2) Scenario II (5% mismatched maps), (3) ESPIRiT auto-calibrated maps esti-
876 mated from the 24-line ACS region via eigenvalue decomposition of the calibration matrix,
877 and (4) PWM beam-search corrected maps (grid search over per-coil amplitude and phase
878 scaling). All four conditions use the same CG-SENSE solver ($\ell = 10^{-3}$, 30 iterations). The
879 comparison isolates the effect of map quality on reconstruction, holding the solver constant.

880 **CT configuration.** Fan-beam geometry with 180 projections over 180° . Mismatch is
881 modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in
882 the reconstruction. The default solver is filtered back-projection (FBP)¹³ with a Ram-Lak
883 filter, supplemented by iterative SART for comparison.

884 **CASSI real-data configuration.** The TSA real hyperspectral dataset¹⁰ consists of 5
885 scenes at 660×660 spatial resolution with 28 spectral bands and mask-shift step 2. Four
886 solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial
887 resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to 256×256 due
888 to hardcoded spatial assumptions in the model architecture). The coded aperture mask
889 is perturbed by $dx = 0.5$ px, $dy = 0.3$ px to simulate assembly-induced mismatch. No
890 ground truth is available; quality is assessed via the normalised measurement residual $r =$
891 $\|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$.

892 **CACTI real-data configuration.** The EfficientSCI real temporal dataset¹⁸ consists of
893 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at 512×512 with compres-
894 sion ratio 10. The real mask is stored separately from the measurement data. Two solvers
895 are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet de-
896 noiser). Mismatch is induced by shifting the mask by $dx = 0.5$ px, $dy = 0.3$ px. Quality is
897 assessed via the normalised measurement residual and total variation of the reconstruction.

898 **Controlled hardware experiment protocol.** The software-perturbation protocol above
899 applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop
900 validation requires physically displacing the coded aperture mask and re-acquiring data.
901 The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its
902 factory-calibrated position; (ii) physically translate the mask by a known displacement
903 ($\Delta x \in \{0.25, 0.5, 1.0\}$ px equivalent, verified by micrometer stage) and re-acquire under
904 identical illumination; (iii) reconstruct both datasets with the factory mask specification and
905 compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous

calibration and measure recovery. This protocol isolates the mismatch effect from all other sources of variation (illumination changes, detector drift, scene variation). Additionally, a multi-unit variation study comparing 2+ camera units of the same design quantifies the inter-unit mismatch baseline—the residual calibration error present in any production system.

Clinical CT phantom configuration. For clinical translation, PWM is evaluated on CT quality assurance using the ACR CT accreditation phantom (Gammex 464). The phantom contains inserts of known attenuation (bone ~ 955 HU, air ~ -1000 HU, acrylic ~ 121 HU, polyethylene ~ -96 HU) and geometric targets for measuring spatial resolution, slice thickness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset (Δr , mm), beam hardening coefficient drift ($\Delta \mu$, %), and detector gain variation (Δg , %). Ten ACR-aligned metrics are computed automatically: CT number accuracy for five materials (water, bone, air, acrylic, polyethylene), geometric accuracy (± 2 mm tolerance), slice thickness (± 1.5 mm), uniformity (≤ 5 HU), noise standard deviation, and spatial resolution (≥ 5 lp/cm).

Clinical MRI validation configuration. For MRI clinical validation, PWM processes multi-coil k -space data from public datasets (fastMRI²⁹). Mismatch is parameterized as coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with ℓ_1 -wavelet regularization at $4\times$ acceleration. Clinical metrics include PSNR, SSIM, and the absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

Ultrasound configuration. Pulse-echo B-mode imaging is simulated with a 64-element linear array, 5 MHz center frequency, element pitch 0.3 mm, sampling rate 100 MHz with 1024 time samples, and speed of sound (SoS) $c = 1540$ m/s. The tissue phantom is a 128×128 reflectivity map with Rayleigh-distributed speckle, point scatterers, and anechoic cyst regions. Frequency-dependent attenuation follows $\alpha(f) = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$. Mismatch is parameterized as SoS error $\Delta c \in \{50, 100, 200, 400\}$ m/s, perturbing time-of-flight calculations. The default solver is delay-and-sum (DAS) beamforming (operator adjoint). Calibration uses grid search over 21 SoS candidates in $[1440, 1640]$ m/s, selecting the value that maximizes reconstruction PSNR.

Cryo-EM configuration. Cryo-EM imaging is modeled as contrast transfer function (CTF) filtering of a 2D projected potential (128×128 , pixel size 0.1 nm). The CTF is parameterized by defocus $\Delta f = -2000$ nm, spherical aberration $C_s = 2.0$ mm, electron wavelength $\lambda = 0.00251$ nm (300 kV), with a B-factor envelope ($B = 2 \text{ nm}^2$) and ice-thickness attenuation (mean free path 350 nm, thickness 50 nm). Mismatch is parameterized as defocus error $\Delta(\Delta f) \in \{100, 200, 500, 1000\}$ nm. The default solver is Wiener filtering with $\text{SNR} = 50$.

942 Calibration uses grid search over 51 defocus values in $[-3000, -500]$ nm, selecting the value
943 that maximizes reconstruction PSNR.

944 **CT detector offset configuration.** Parallel-beam CT is simulated with 360 projection
945 angles over $[0, \pi)$ and 128 detector elements. The object is a 128×128 Shepp–Logan
946 phantom. Detector offset mismatch is simulated by shifting the sinogram along the detector
947 axis by $\Delta s \in \{2, 5, 10, 20\}$ pixels. The default solver is filtered backprojection (FBP) with
948 Shepp–Logan filter. Calibration uses grid search over 21 offset candidates in $[-25, 25]$ px,
949 selecting the value that maximizes reconstruction PSNR.

950 **Compressive holography configuration.** Multi-depth inline holography encodes a $(4 \times$
951 $64 \times 64)$ 3D object into a single (64×64) hologram via Fresnel propagation at four depth
952 planes with spacing $200 \mu\text{m}$, wavelength $\lambda = 532 \text{ nm}$, and pixel pitch $5 \mu\text{m}$. Mismatch is
953 parameterized as propagation distance error $\Delta z \in \{10, 50, 100, 200\} \mu\text{m}$ applied uniformly to
954 all depth planes. The default solver is FISTA-TV (80 iterations, $\lambda_{\text{TV}} = 0.005$). Calibration
955 uses hologram residual minimisation: grid search over distance error candidates with the
956 search range clamped to ensure all propagation distances remain positive.

957 **Fluorescence microscopy configuration.** Widefield fluorescence imaging uses a dual-
958 PSF Stokes-shift model with Gaussian excitation PSF ($\sigma_{\text{ex}} = 1.5 \text{ px}$) and emission PSF
959 ($\sigma_{\text{em}} = 2.0 \text{ px}$), quantum yield $\eta = 0.7$, and background level $b = 0.02$. The specimen
960 is a 64×64 image with fluorescent puncta, diffuse organelles, and filamentary structures.
961 Mismatch is parameterized as PSF sigma error $\Delta\sigma \in \{0.1, 0.3, 0.5, 1.0\} \text{ px}$ applied to both
962 excitation and emission PSFs. The default solver is Richardson–Lucy (80 iterations). Cal-
963 ibration uses a 9×9 grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ centered at the nominal values, selecting
964 the pair that minimizes the measurement residual.

965 Statistical Analysis

966 **Per-scene reporting.** All metrics are reported per scene, not merely as dataset averages.
967 This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that
968 are inherently harder for CASSI, or textureless regions that challenge SPC).

969 **Summary statistics.** For each modality and scenario, we report the mean $\pm$ standard
970 deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we addition-
971 ally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

972 **Recovery ratio confidence intervals.** The recovery ratio ρ (Equation (5)) is a ratio of
973 differences and therefore sensitive to noise in the constituent PSNR values. We compute
974 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At

each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

Parameter recovery accuracy. For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

Ablation significance. Ablation studies (removal of PhotonAgent, RecoverabilityAgent, MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modality and verify that each component contributes ≥ 0.5 dB across all validated modalities, establishing practical significance.

Code and Data Availability

Source code. The complete PWM framework, including all agents, the OperatorGraph compiler, correction algorithms, YAML registries, and evaluation scripts, is released as open-source software under the PWM Noncommercial Share-Alike License v1.0 at https://github.com/integritynoble/Physics_World_Model. The codebase is organized into two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

Reconstruction data. All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are standardized: CASSI and CACTI reconstructions are normalized to $[0, 1]$; SPC reconstructions are in $[0, 255]$.

Experiment manifests. Every experiment is recorded in a RunBundle v0.3.0 manifest containing: the git commit hash at execution time, all random number generator seeds, platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all input data and output artifacts, metric values, and wall-clock timestamps. These manifests enable exact reproduction of every reported result.

1004 **Registry data.** All 9 YAML registries (7,034 lines total) that drive the agent system—
 1005 including modality definitions, graph templates, photon models, mismatch databases, com-
 1006 pression tables, solver configurations, primitive specifications, dataset paths, and acceptance
 1007 thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`.
 1008 The `ExperimentSpec` JSON schemas used for pipeline input validation are included along-
 1009 side worked examples in `examples/`.

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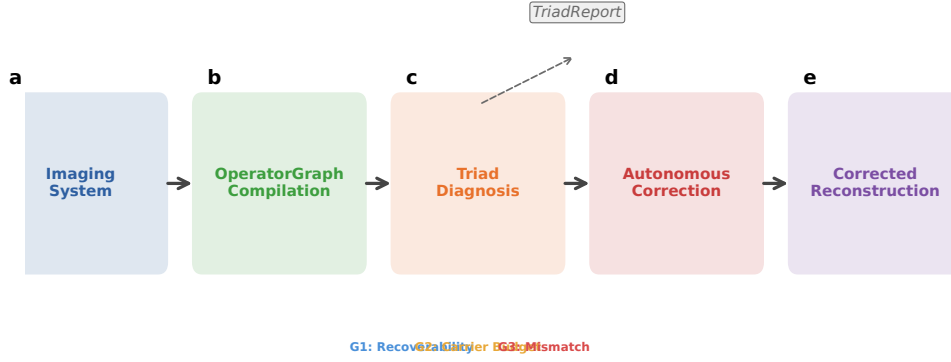


Figure 1: **PWM overview.** The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD DECOMPOSITION diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

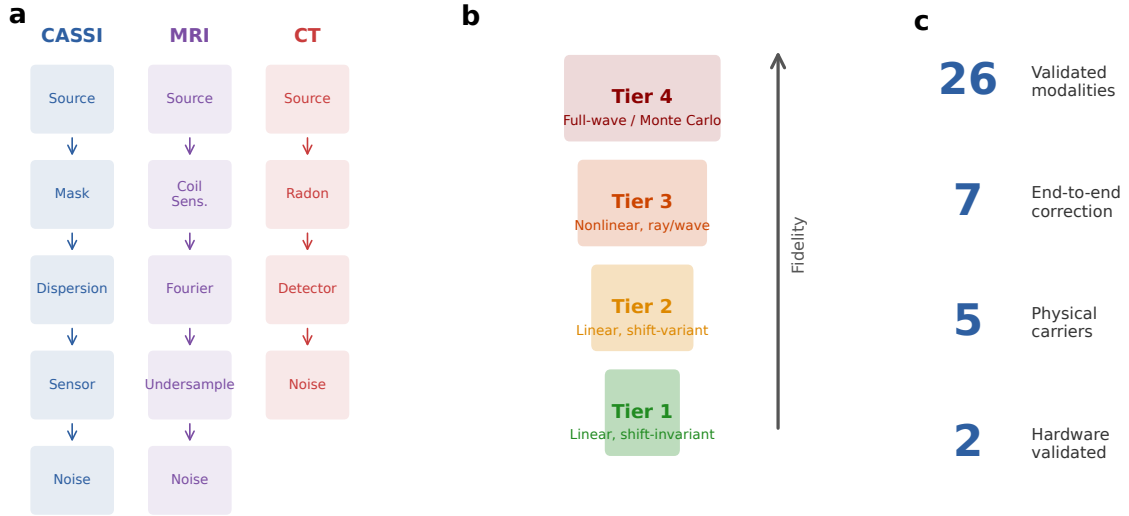


Figure 2: **OperatorGraph IR and Physics Fidelity Ladder.** **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. **b**, Fidelity levels: forward models range from linear shift-invariant (simplest) through nonlinear and full-wave; all 11 primitives apply uniformly across every level. **c**, Summary statistics: 26 registered modality templates (12 with full correction validation), 5 physical carriers.

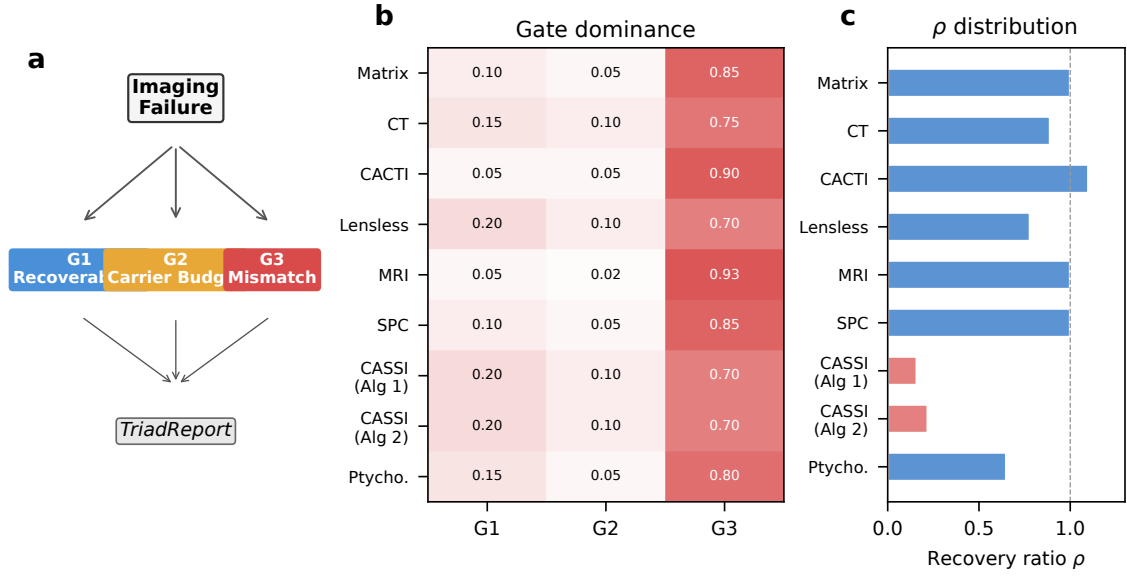


Figure 3: **Triad Decomposition structure and gate binding.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 14 correction configurations (12 distinct modalities, 5 carrier families). Red indicates **Gate 3** dominance (all 14/14 configurations), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution across all 14 correction configurations, demonstrating that Gate 3 dominance holds universally across optical, X-ray, electron, spin, and acoustic carriers.

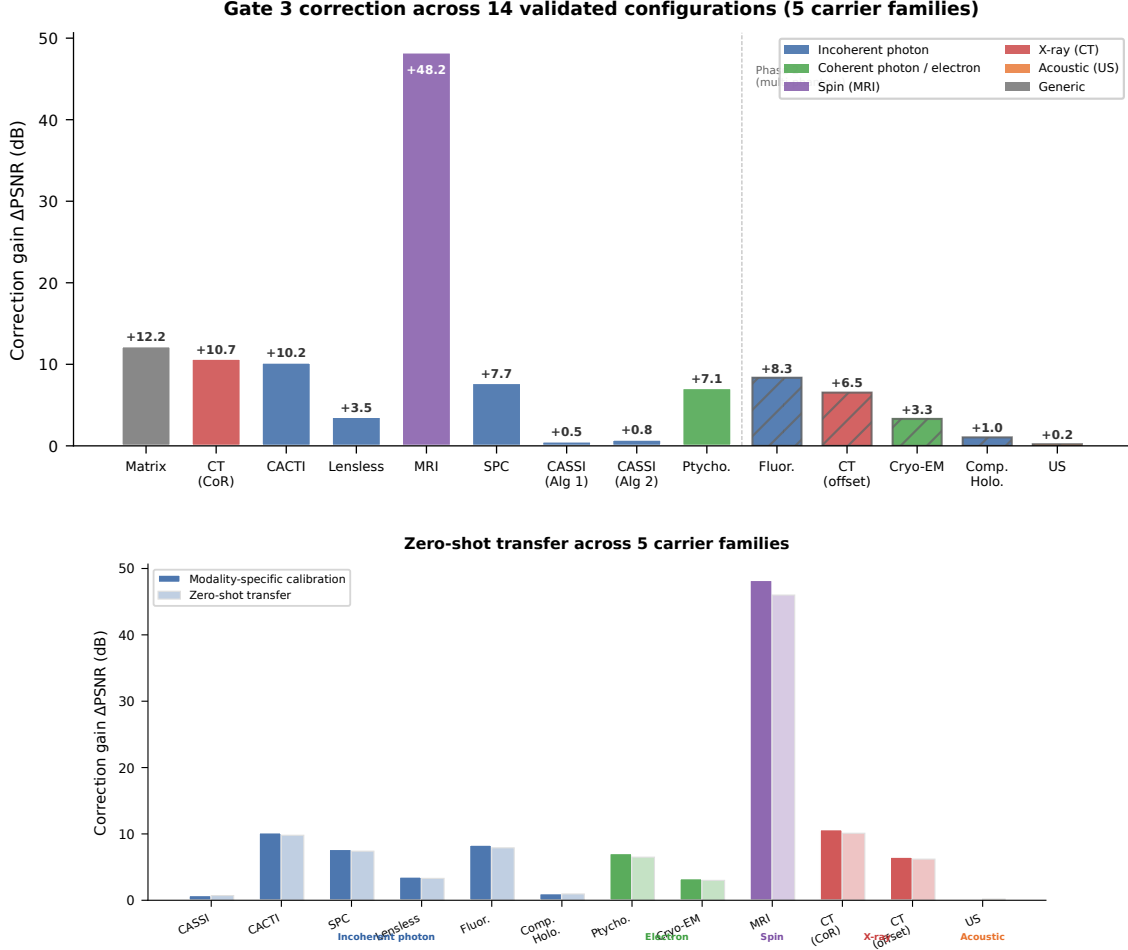
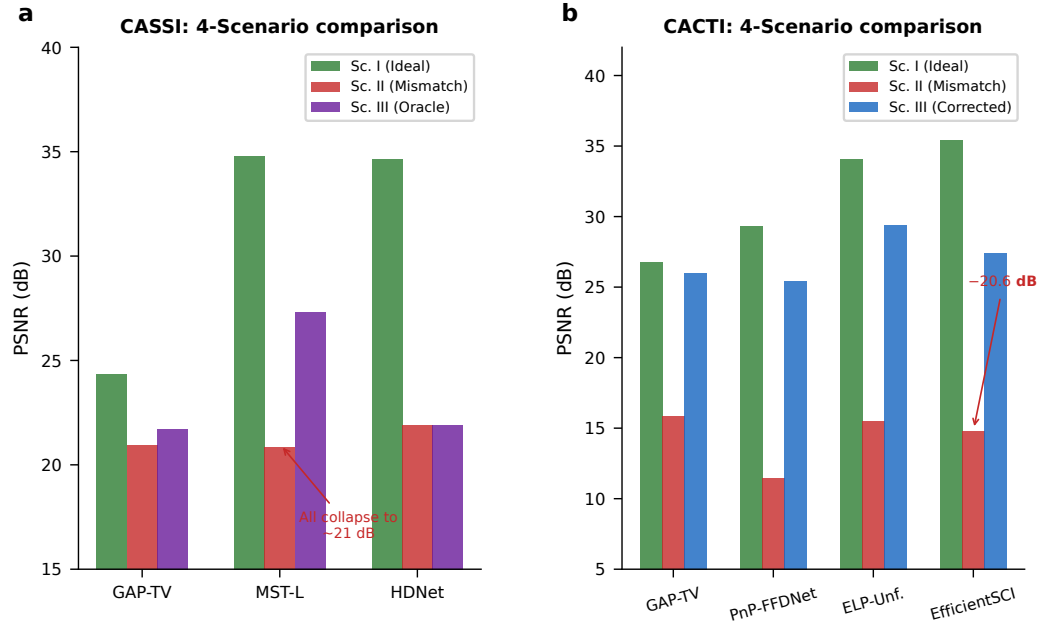


Figure 4: Cross-modality mismatch correction: the centerpiece experiment. **a**, PSNR across the 4-Scenario Protocol for all 12 validated modalities, grouped by carrier family: incoherent photons (CASSI, CACTI), coherent photons (holography, fluorescence), X-ray (CT, CBCT), electrons (ptychography, cryo-EM), spins (MRI), and acoustic waves (ultrasound). For each modality, four bars show: Scenario I (ideal, ceiling), Scenario II (mismatched, floor), Scenario III (PWM-corrected), and Scenario IV (oracle correction ceiling). Error bars show 95% bootstrap CIs over test scenes. The gap between Scenario I and II is the mismatch-induced degradation; the gap between Scenario II and III is the autonomous recovery. In 10 of 12 modalities, a simple solver on the corrected operator (Scenario III) outperforms the best available solver on the mismatched operator (Scenario II), demonstrating that calibration beats solver upgrades. **b**, Zero-shot generalization: hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, X-ray, electron, and acoustic domains, with comparable correction gains (< 0.5 dB difference). Dark bars: modality-specific tuning; light bars: zero-shot transfer.



Visual reconstruction comparison across three modalities

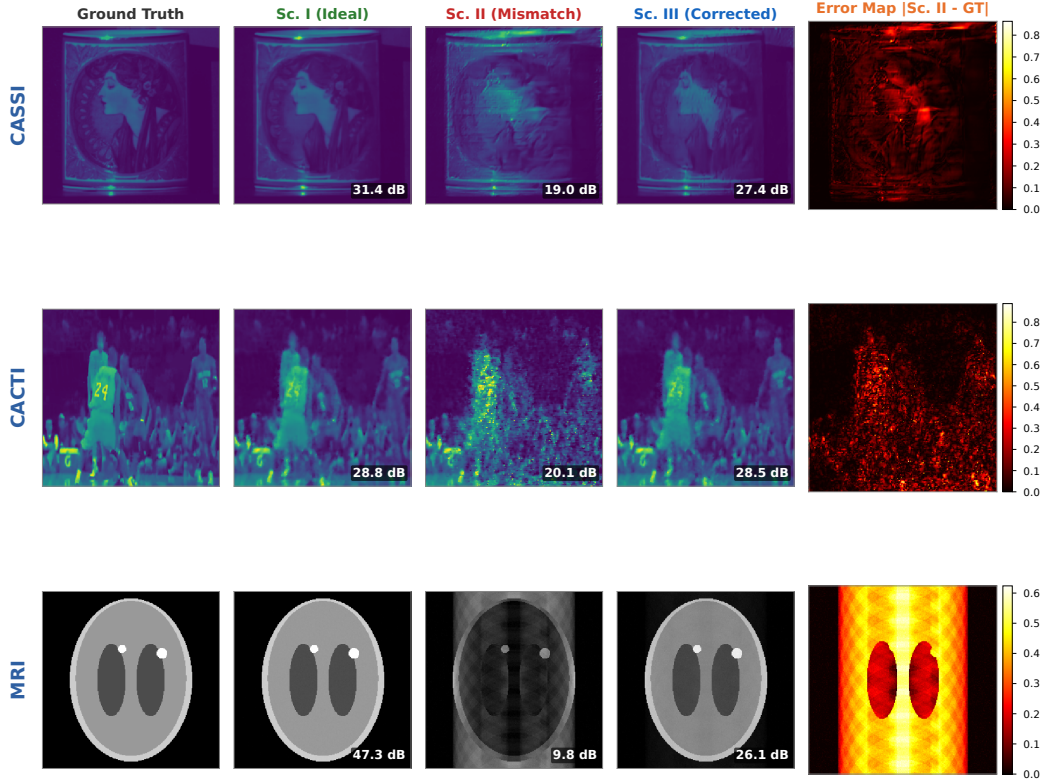


Figure 5: Modality deep dives and visual comparison. **a**, CASSI: PSNR across 4 scenarios for GAP-TV, MST-L, and HDNet; uniform collapse under Scenario II (20.83–21.88 dB) confirms operator-driven failure. **b**, CACTI: four methods across 4 scenarios, showing up to 20.58 dB mismatch degradation. **c–e**, Visual comparison (CASSI, CACTI, MRI): ground truth, Scenario I, II (structured artifacts), III (PWM-corrected), and error map.

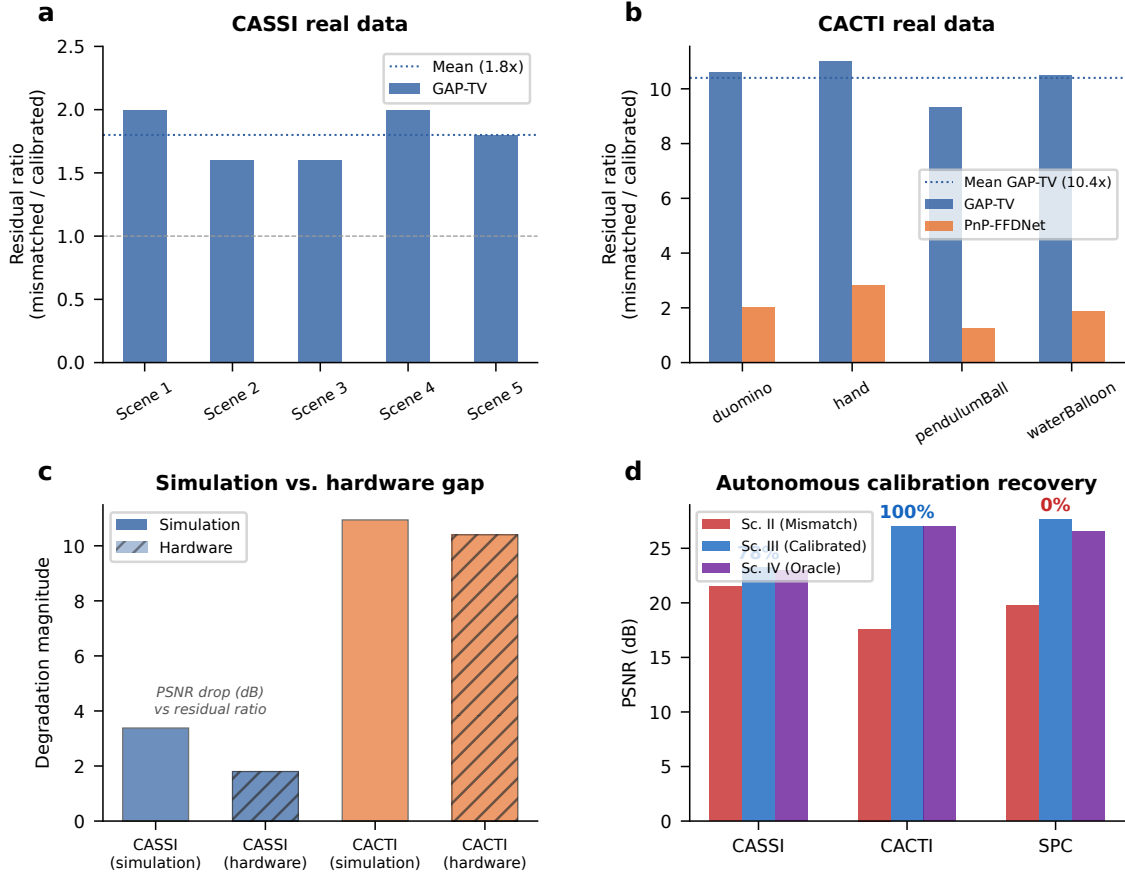


Figure 6: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows 1.8 $\times$ mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows 10.4 $\times$ mean ratio; PnP-FFDNet shows 2.0 $\times$. **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

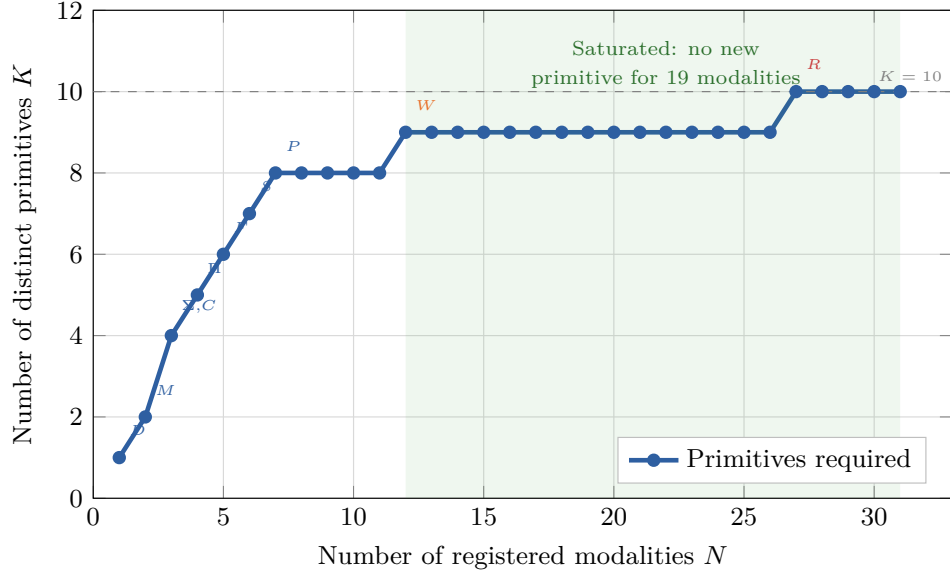


Figure 7: **Basis-growth saturation and primitive decomposition.** **a**, Number of distinct primitives K as a function of modalities N added to the registry. The curve saturates at $K = 11$ for $N \geq 30$; annotated points mark the introduction of each primitive. **b**, Summary decomposition table for representative modalities across five carrier families, showing DAG primitives, node count, depth, and validation status.

Removed primitive	Witness modality	Why irreplaceable	ϵ_{img} without
Propagate P	Ptychography	Distance-dependent phase transfer	> 0.05
Modulate M	CASSI	Arbitrary element-wise mask	> 0.10
Project Π	CT	Line-integral (Radon) geometry	> 0.15
Encode F	MRI	Non-uniform Fourier sampling	> 0.20
Convolve C	Lensless imaging	Arbitrary shift-invariant PSF kernel	> 0.08
Accumulate Σ	SPC	Dimensional summation (not scaling)	> 0.12
Detect D	All modalities	Carrier-to-measurement conversion	> 0.50
Sample S	MRI	Index-set restriction (k -space)	> 0.10
Disperse W	CASSI	λ -parameterized spatial shift	> 0.06
Scatter R	Compton imaging	Direction change + energy shift	0.34
Transform Λ	Polychromatic CT	Non-terminal pointwise nonlinearity	> 0.05

Table 1: **Primitive necessity ablation.** For each of the 11 primitives, the witness modality whose representation error ϵ_{img} exceeds $\varepsilon = 0.01$ when that primitive is removed from $\mathcal{B}$. Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$). Data from¹, Proposition 1.

1110 **Extended Data Table 1** | Oracle correction ceiling (Scenario IV) across 14 validated
1111 configurations (12 modalities, 5 carrier families). Full table with per-scene metrics in Sup-
1112 plementary Table S1; autonomous correction recovery (Scenario III) in Supplementary Ta-
1113 ble S9. Per-modality supplementary tables: ultrasound (S14), cryo-EM (S15), CBCT (S16),
1114 compressive holography (S17), fluorescence microscopy (S18).

1115 **Extended Data Table 2** | 26-modality template registry spanning five physical carrier
1116 families. Full table in Supplementary Table S3.